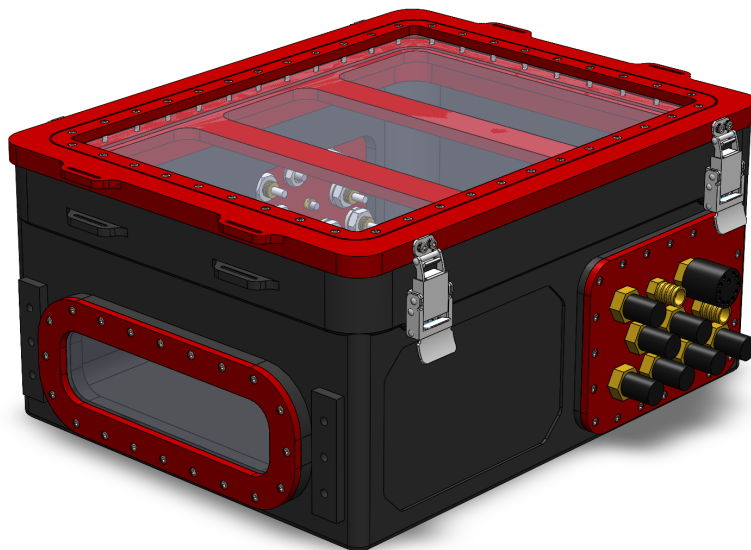




*Cornell University Autonomous Underwater
Vehicle Team*

Fall 2024

Orion UHPV



Technical Report

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1 Abstract

The UHPV (Upper Hull Pressure Vessel) is the primary pressure vessel of the AUV. It encloses the circuit boards, the Jetson, and the forward-facing camera, and needs to be securely mounted to the frame. These elements inside the UHPV are able to communicate with other systems on the sub (such as the thrusters, hydrophones, and the DVL) using SEACON cables, which are connected to the UHPV using SEACON bulkheads. A ball valve is used to depressurize the UHPV to aid with assembly and to help engage the bore seal.

The Orion UHPV is another iteration of the rectangular (or "lunchbox") style UHPV, building off of Sirius's more novel design. Orion's design seeks to optimize the weight of the lunchbox-style design and to improve the manufacturing and ease-of-assembly of the bore seal between the lid and the hull.

2 Design Requirements

2.1 Constraints

- Safely enclose the circuit boards and the Jetson, preventing any water damage
- House all of the SEACON bulkheads required to communicate with all external components of the AUV
- Be securely mounted to the frame
- Contain the forwards-facing (ZED) camera, killswitch, and depth sensor

2.2 Objectives

- Minimize the weight (and size) of the UHPV
- Make the lid bore seal easy to remove without compromising seal quality
- Accelerate the UHPV project timeline so that Orion can be completed faster, allowing for more software testing
- Increase the accessibility of components that are commonly disassembled/reassembled

3 Previous Designs

3.1 Rectangular ("Lunchbox") UHPVs

Throughout most of CUAUV's history, the main pressure vessels of each sub have traditionally been cylindrically-shaped, primarily due to their ease of design and manufacturing. More recently, however, we have begun to experiment with rectangular-shaped (commonly

referred to as "lunchbox") UHPVs, as they provide certain benefits that cylindrical UHPVs do not. For one, our PCB boards are rectangular and can be packed most efficiently inside a rectangular design, which reduces the total volume required for the UHPV. The lid/hull design also makes accessing the boards significantly easier than with the midcap/endcap design on the cylindrical UHPVs, which is beneficial in time-sensitive situations (which are common during our Robosub competition). So far, we have created two rectangular UHPVs: Pollux (2017-2019) and Sirius (2024-present).

3.1.1 Pollux (2017-2019)

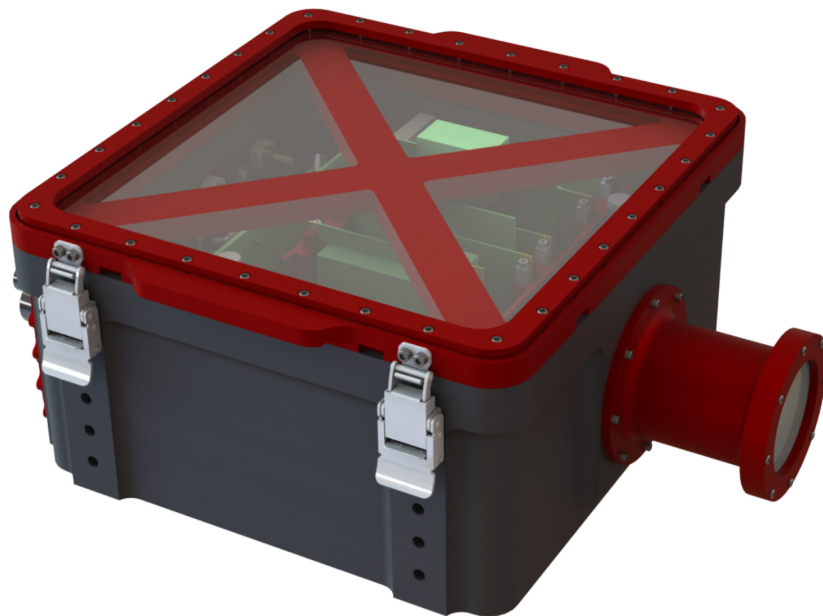


Figure 1: Pollux's UHPV

The Pollux UHPV was the first attempt at a rectangular design. Pollux was designed to be a minisub, so the overall dimensions of the UHPV are smaller to condense the overall size of the sub. The UHPV is mounted to the frame on the port and starboard sides of the hull, and the SEACON bulkheads are all condensed to a single panel on the aft face of the hull. AUVs previous to Pollux tended to be too positively buoyant and required additional weights to be added to the frame, so the smaller internal volume of the UHPV helped to reduce its buoyancy. The forwards-facing camera (a UI camera) also has an additional extending piece (appearing similar to other camera enclosures) to accommodate it the proper space needed.

Once fully integrated, the interior of the UHPV was extremely compact, making it difficult to access the boards and debug the electrical components of the AUV. The single

SEACON panel also caused much of the weight of the UHPV to be condensed to the aft, which in turn made Pollux unbalanced. Pollux's UHPV was machined entirely in-house (via subtractive manufacturing), and although it was particularly difficult to machine the hull due to the depth-of-cut required, the non-circular bore seal was successfully manufactured. Overall, Pollux's design was a successful proof-of-concept for the rectangular UHPV, and highlighted its potential as an effective option for a mainsub.

3.1.2 Sirius (2024-Present)



Figure 2: Sirius's UHPV

The Sirius UHPV built upon the lessons learned with Pollux's design and scaled the UHPV up to serve as a mainsub. Rather than using standard subtractive manufacturing, Sirius's hull was made from welding together plates for each side. To reduce the number of separate enclosures on Sirius, the killswitch, IMU (the GX sensor), and forwards camera (ZED camera) were all moved to be housed inside of the UHPV. The UHPV is mounted to the frame on the fore/aft sides, and rather than a single aft SEACON panel, the Sirius UHPV has two panels on the port/starboard sides. To save on weight, cutouts in the hull were made on the fore/aft sides as well as on the bottom, and an acrylic viewport allows the stereoscopic ZED camera to be nested inside of the hull while having an unobstructed view.

Due to the difficulties of welding aluminium plates, there was some warping of the hull of the UHPV, including the collar piece (which forms the bore seal with the lid). This caused the lid to not fit smoothly into the collar, requiring a significant amount of sanding

and a tight fit between the lid and the collar. This required the use of a mallet to remove the lid, making the bore seal a bit inconvenient to work with.

On the whole, however, the Sirius UHPV was a successful implementation of the rectangular UHPV design. Separating the SEACON bulkheads onto two panels helped the UHPV stay significantly more balanced, and Sirius was quite controllable in the water, easily capable of completing barrel rolls and other movement requirements. Incorporating the killswitch and forwards camera into the UHPV design also helped to save on space and weight, and having the killswitch next to the tether bulkhead was extremely convenient when preparing autonomous runs at competition. The more compact design decreased the buoyant force supplied by the UHPV, but unlike with Pollux, Sirius was already quite negatively-buoyant. Thus, Sirius required a significant amount of extra trim to overcome its 15 pounds of negative buoyancy.

3.2 Cylindrical UHPVs

Outside of Sirius and Pollux, all of our prior AUVs have featured a cylindrical UHPV. These designs typically consist of an aluminium midcap (which houses the SEACON bulkheads and connects to the frame of the sub) and one or two acrylic endcaps (which encompass the electrical boards, Jetson, and other components). Although less compact, these designs tend to be more positively buoyant, and are typically lighter than a rectangular UHPV (due to more acrylic and less aluminum being used in the design). Midcaps are typically re-used for multiple AUVs, since they are expensive and difficult to manufacture.

3.2.1 Leviathan (2020-2023)

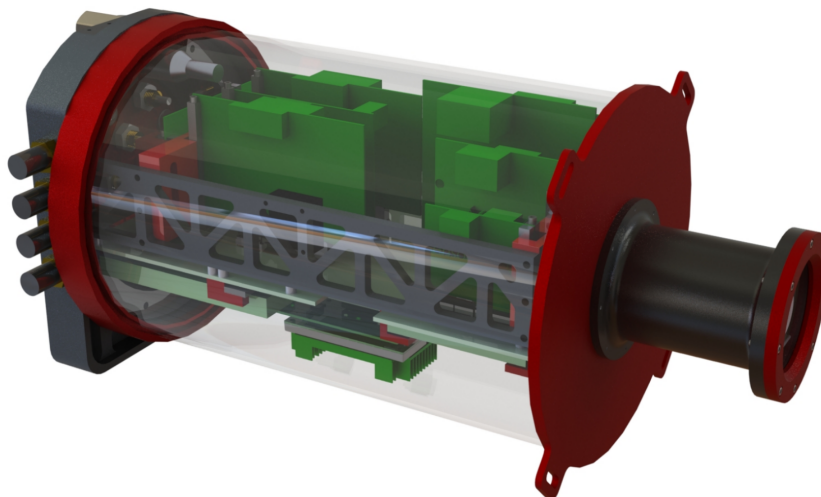


Figure 3: Leviathan's UHPV

Leviathan is a minisub, so its UHPV is designed to be simple and compact. It features a single endcap with a camera enclosure-like extension to accommodate a forward-facing UI camera. The midcap is also quite minimal, with its SEACON bulkheads primarily on the aft face, with a few more lining the rim along its sides. Due to the asymmetry of the design, much of the UHPV's weight is concentrated towards the aft, and its generated buoyant force is more towards the front, causing the Leviathan UHPV to be quite unbalanced. Its small size also led to the electrical boards being quite cramped, which made them more difficult to access and maintenance.

3.2.2 Polaris (2022-2023)

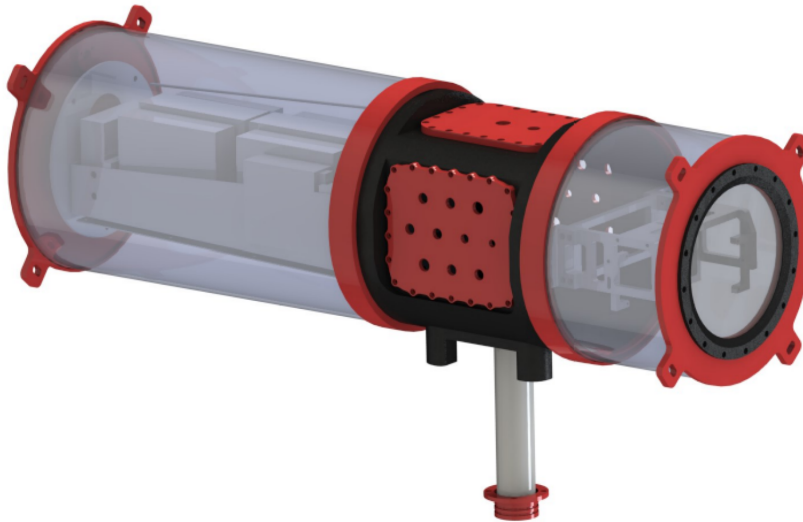


Figure 4: Polaris's UHPV

Polaris sports the most recent cylindrical UHPV design, opting for the more typical two-endcap design. The midcap was re-designed and outsourced, and has most of its SEACON bulkheads on its port and starboard plates. The fore endcap has a viewport and contains the Jetson and a UI camera, while the aft endcap houses the electrical boards. Unique to most designs, Polaris initially had a "stem" connection to its downwards-facing camera (a watertight connection for wires), but this was later removed and a separate camera enclosure was created. As Polaris aged, its bore seal became increasingly difficult to engage, requiring two or more people to install the endcaps. This was likely due to burring along the sealing surfaces and an inadequate chamfer to guide the o-rings during installation.

4 High Level Description

The Orion UHPV design is another rectangular ("lunchbox") UHPV which is quite similar to Sirius. Since Sirius (and other rectangular UHPVs) are quite heavy, Orion seeks to be as lightweight as possible, allowing more weight to be allocated to other areas of the AUV (such as the manipulator). In addition, Sirius's design experienced numerous delays due to leaking and issues with manufacturing the bore seal, so Orion's UHPV also aims to mitigate these potential issues to accelerate the project timeline (and allow Orion to be more thoroughly tested in the water prior to competition). Unlike with Sirius, the Orion UHPV will be made entirely from subtractive manufacturing, including the hull (which will be outsourced to Xometry).

4.1 Lid Assembly

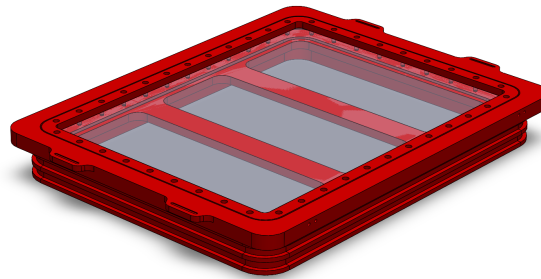


Figure 5: The full lid assembly, consisting of the lid, acrylic sheet, and the lid flange.

The lid assembly consists of the lid itself (the large outer part), an acrylic sheet, and a lid flange (which fastens the acrylic sheet to the lid).

The lid serves as the male side of the bore seal, fitted with two 384 o-rings that slot into the hull when the lid is closed. In Sirius and prior designs, 200-series o-rings were used for this bore seal, which have a smaller cross-sectional diameter (0.139") than the 300-series o-rings (0.210"). Using a thicker o-ring adds additional tolerance to the various dimensions of the bore seal, so switching to the 300-series should help with machining the lid and the hull. Additionally, as a deviation from the Parker O-Ring Handbook, an additional 20 thou was removed from the outer edge (radially) of the lid in order to increase the clearance gap between the lid and the hull. This is to reduce the issues experienced with Sirius's design where the lid did not slot into the collar after machining, requiring extensive sanding. For additional information regarding this bore seal and other o-ring seals, see Appendix C.

The acrylic sheet is 1/4" thick and is supported by the lid with two cross-beams. The acrylic sheet is face sealed to the lid with a 279 o-ring. Since the corners of the lid tend to fit the tightest with the hull, Orion's lid features split handles to make it easier to remove the lid. The lid flange is 1/8" thick and will be outsourced to Datron.

4.2 Hull



Figure 6: The hull of Orion's UHPV

The hull of Orion's UHPV is quite similar to Sirius's design; however, it has been designed for subtractive manufacturing rather than welding. The overall dimensions of the hull have been slightly reduced from Sirius, having a width of 11.7", length of 15.55", and a height of 6.95".

The top of the hull acts as the female side of the bore seal, and follows all of the dimensions as dictated by the Parker O-Ring Handbook. Unlike on previous subs, a 20 degree chamfer was added on the inner edge which receives the o-rings when the lid is closed (as recommended by the handbook), which should make it easier to close the lid without the extrusion or nibbling of the o-rings. For more information regarding the bore seal and other o-ring seals, see Appendix C.

The mounting holes for the frame are still located along the fore and aft sides of the hull, as well as underneath the UHPV. The depth sensor has also been moved from the base of the UHPV to the killswitch/tether panel.

4.3 SEACON Panels

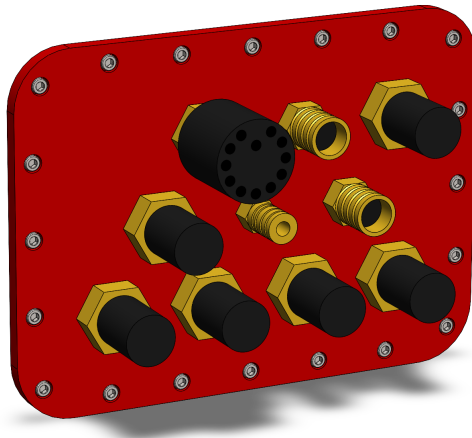


Figure 7: The starboard SEACON panel, including the SEACON bulkheads.

The port and starboard SEACON panels are structured similarly to those on Sirius. To condense the SEACON panels down, the bulkheads are arranged in a more square pattern (rows of 3-3-4 rather than the two rows of five on Sirius), which has helped increase the weight savings possible on the hull. No bulkheads were changed between Sirius's and Orion's designs, so the electrical integration of both subs will be nearly identical. Both panels feature an o-ring groove on their inner surface to fit a 249 o-ring.

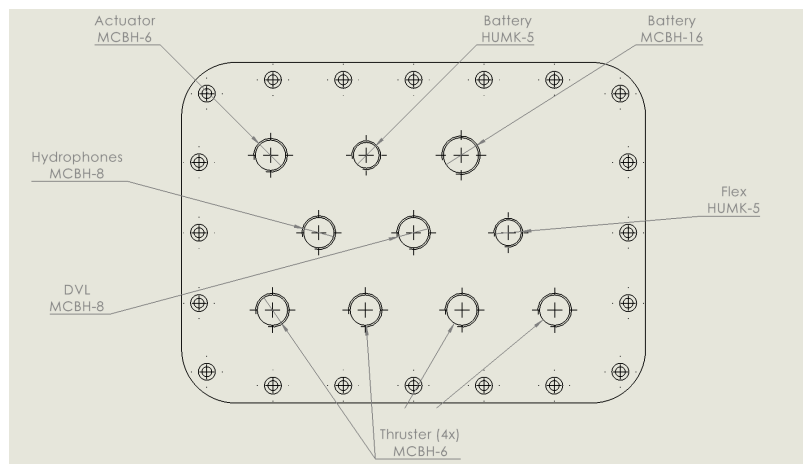


Figure 8: A layout of the SEACON bulkheads (Port)

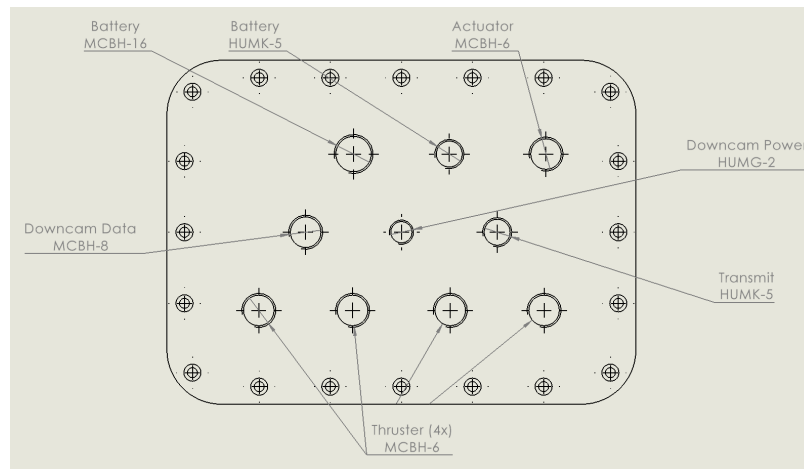


Figure 9: A layout of the SEACON bulkheads (Starboard)

4.4 Killswitch (Aft) and ZED Camera (Fore) Panels

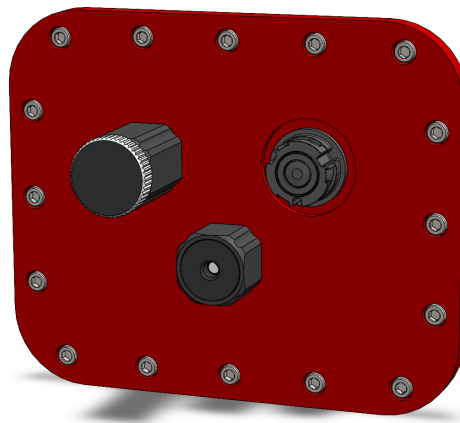


Figure 10: The killswitch panel, consisting of the killswitch, the new Binder 770 bulkhead, and the Bar02 depth sensor.

The killswitch (aft) panel has some major changes compared to Sirius. The tether bulkhead on Orion has been changed from the more standard MCBH-8 bulkhead to a Binder 770 bulkhead, which interfaces directly with BlueRobotics Fathom tether without the need for splicing. Since we have had connection issues with the ethernet-spliced tethers, this change should help to make our tethered connections more consistent and reliable at competition. The killswitch itself is the standard BlueRobotics Switch (BR-100433) which is relatively cheap and consistent part that is easy to replace if it happens to break. Additionally, the

depth sensor for Orion has been moved to this panel. We have switched from the older and larger depth sensor to the BlueRobotics Bar02 Ultra High Resolution sensor, which is not only smaller, but provides highly-accurate depth information (with a resolution below 0.5 mm). We have had issues in the past with specialized sensors from other companies being discontinued or requiring slow quoting processes, so switching to a BlueRobotics part should be a more reliable alternative. The killswitch panel is sealed with a 235 o-ring.

The ZED camera (fore) panel is largely unchanged from Sirius's design. The acrylic viewport is 1/8" thick, and is sealed with a 255 o-ring. Unlike the other panels, the ZED panel requires the o-ring groove to be machined directly into the hull.

5 Manufacturing

The manufacturing of the UHPV is split between parts done in-house, parts outsourced to Datron, and the hull which is being outsourced to Xometry.

The current plan is to machine the lid on the CNC Haas in the Emerson Machine Shop, since it is a larger part (that cannot be outsourced to Datron) but can be accomplished in a reasonable amount of time. This also allows us to more directly control the tolerance of the bore seal, which may prove to be useful if adjustments need to be made. The acrylic sheets for both the lid and the ZED camera viewport can also be laser cut by us.

The panels and the lid flange are all 1/4" thick, so they can be outsourced to Datron. Since it would take us a significant amount of time to machine all of the holes and o-ring grooves, it makes sense to focus our machining efforts to other parts on the sub. If the need were to arise, however, the panels could be re-machined or altered using the CNC, which could be useful if the UHPV experiences repeated issues during leak testing.

5.1 Xometry

Due to the issues that arose with welding Sirius, the limitations of the CNC Haas, and an emphasis on an accelerated timeline, the hull of the Orion UHPV will be outsourced to Xometry. We considered a few different companies that could machine the hull, but ultimately Xometry offered a significantly cheaper quote with reasonable turn-around time (2-3 weeks).

The total cost of the hull will be around \$2,500, which is a significant cost but will be a worthy investment given the time and effort it would otherwise take to machine the hull. By submitting drawings with our order, we can precisely tolerance the o-ring seals on the hull and have all of the holes tapped. To ensure the hull is machined correctly, we are also ordering a CMM inspection of the hull (with a report of all the critical dimensions we specify) for a small additional fee (\$50).

6 Current Status

The Orion UHPV has just finished its design phase. Parts that are planned to be outsourced will be sent out during winter break so that they are ready to be assembled during the first couple of weeks into the spring semester. To help inform Orion's design, Sirius's UHPV designer (Grace Shirota) was consulted to determine the most important changes to focus on and lessons learned from Sirius. Special thanks to her for all of her help!

7 Future Improvements and Reflection of Process

The overall design strategy behind Orion's UHPV was to make small, low-risk changes to Sirius's largely-effective design in order to accelerate Orion's overall manufacturing and integration timeline. One of the main issues of our previous design cycle was the extensive delays that set back both the UHPV and other enclosures/parts, greatly limiting the amount of time Sirius could be tested in the water leading up to competition. Although I am very supportive of more experimental projects, it is important to consider what types of issues may arise with a novel design and plan accordingly, expecting most things to take longer than expected.

Although many of the changes made between Sirius and Orion are subtle, I believe they are important for optimizing the rectangular UHPV design and more fully determining its capabilities moving forward. I spent a significant amount of time reading the Parker O-Ring Handbook and other resources to see how the bore seal (and o-ring seals in general) could be better implemented into our designs. I would highly recommend checking out Appendix C for more information, since some of the extra details I describe there may be useful in future designs for the UHPV and other enclosures, even if not immediately applicable to Orion.

Moving forwards, I think that there are some directions that the rectangular UHPV design can be taken. If supported by the electrical subteam, I believe adding one or more LCD screens could be a productive project with the potential to win design awards at competition and potentially increase our point potential. An LCD screen could convey to a diver (or us) what the sub is aiming to do during autonomous runs, either by text or by picture (using glyphs similar to those on the scoring sheets). It would also be a way to visibly show our inter-sub communication being used to influence our competition strategy (such as sending one sub to a certain location based off of the pinger), increasing the number of points we receive for that inter-sub communication (1,000 max). There may also be ways to incorporate more acrylic into the rectangular UHPV design (similar to the endcaps in the cylindrical UHPVs) to reduce weight.

7.1 Recommended Timeline for UHPV Design

The timeline for the UHPV heavily dictates the timeline for its AUV; if the UHPV is delayed, then the entire project timeline can be set back by weeks or months. As such, it is

important that the proper amount of time is budgeted for the UHPV, factoring in delays and anticipating that some setbacks will occur in the process.

Sirius's design was quite novel, and despite being started in July of the previous year, it was still not successfully leak-tested until halfway through the spring semester. Thus, I would recommend that UHPV designs in the future are started during the summer or towards the end of the spring of the previous year. Moving PPDRs and PDRs to occur before the fall semester can help to move the UHPV along and set aside enough time for the MDR and FDR, which require the most CADing and attention-to-detail.

If the UHPV design is not started until the beginning of the fall semester, then I would recommend scaling-back how ambitious that design will be. Although it is important to improve on previous designs, it is much more critical that the UHPV is completed on-time. I did not start Orion before this fall, and I believe my more cautious approach will prove to be useful.

7.2 Rectangular VS Cylindrical UHPVs

Although we have attempted a few rectangular UHPVs (now including Orion), I think it is important to consider the benefits and drawback of each design to determine which design should be used for which AUV moving forwards. I am not under the impression that either design is better in all circumstances, and the pros/cons should be analyzed on a year-by-year basis.

Rectangular UHPVs are more compact, but this has not always been a benefit to each AUV. Pollux was naturally more positively-buoyant, so the smaller volume of the UHPV helped to keep it closer to neutrally-buoyant (thus requiring less trim). Sirius was naturally quite negatively-buoyant, so the UHPV providing less buoyant force meant that lots of additional trim was needed. If our full-sized mainsubs continue to be a similar weight, then this will continue to be a challenge for the rectangular UHPV.

Weight is the other main concern for rectangular UHPVs. A lot more aluminum is required for a rectangular UHPV, and it seems very unlikely that one could be under 20 pounds without introducing major headaches and potential issues. Cylindrical UHPVs have been consistently lighter due to the abundance of acrylic used for the endcaps. Given that Sirius was close to the weight penalties at competition without several enclosures attached, it will continue to be a challenge moving forwards to design a rectangular UHPV for a mainsub that meets weight requirements.

With these two points aside, there are a lot of benefits that the rectangular UHPV provides which cannot be accomplished by its cylindrical counterpart. The electrical boards are much, much easier to access on a rectangular AUV, which has proven very useful at competition (where quick surgery on electrical boards can make or break an autonomous run). The rectangular UHPV has also proven to be more balanced and easier to control in the water, and mounts securely to the frame (which has been an issue on other cylindrical UHPVs, such as Polaris). Although not strictly necessary, the rectangular UHPV also occupies less space, which may help reduce the amount of material needed for the frame.

Overall, I believe that the rectangular UHPV is the more "ideal" design, but its costs come in the form of more difficult manufacturing and more extensive planning required to address the weight and buoyancy issues. Since the technical experience and available machining time of the mechanical subteam fluctuates year-to-year, the best UHPV design for a particular AUV may also change based on the current composition of the team. It also appears that undertaking a rectangular UHPV for a minisub is easier than a mainsub. Prior to a PPDR, I would recommend considering both options thoroughly.

Appendices

A Components To Purchase

A.1 Stock

Component	Stock Size	Qty.
Lid (Part)	1.75" x 12" x 16"	1
All Panels (Being sent to Datron)	0.25" x 12" x 10.5"	1
Hull	N/A (Selected by Xometry)	1

A.2 Hardware

Component	Qty.
4-40 Super Corrosion-Resistant Screws (9/16", 7/16", and 3/8")	4 total packs
6-32 Super Corrosion-Resistant Screws (For the Draw Latches)	1 pack of 50
1/4"-20 Super Corrosion-Resistant Screws (Attachments to the Frame)	2 packs of 10
4x Corrosion-Resistant Draw Latches	4
1/2" Washers (SEACON Bulkheads)	1 pack of 50
7/16" Washers (SEACON Bulkheads)	1 pack of 50
3/8" Belvedere Washers (SEACON Bulkheads)	1 pack of 5
1/2"-20 Nuts (SEACON Bulkheads)	1 pack of 50
7/16"-20 Nuts (SEACON Bulkheads)	1 pack of 100
3/8"-24 Thin Nuts (SEACON Bulkheads)	1 pack of 50

A.3 O-rings

Component	Size	Qty.
Bore Seal (Lid)	384	1 bag of 5
Lid/Acrylic/Flange Face Seal	279	1 bag of 5
Fore (ZED) Panel Face Seal	255	1 bag of 10
SEACON Panel Face Seals	249	1 bag of 10
Aft (Killswitch) Panel Face Seal	235	1 bag of 50

A.4 Other Integrated Items

Component	Qty.
Bar02 Pressure/Depth Sensor (BlueRobotics)	1
Killswitch (BlueRobotics)	2 (1 spare minimum)
Binder 770 Tether Bulkhead (BlueRobotics)	2 (1 spare minimum)
Ball Pressure Valve	1

B Finite Element Analysis

B.1 Fully Assembled Enclosure

As done with Sirius and previous AUVs, the full Orion UHPV enclosure was tested at 15 PSI, which simulates the forces experienced by the UHPV at a depth of 20 feet under the water. Our AUVs have typically operated 5-10 feet under the surface of the water, and the competition pool at Irvine has a maximum depth of 12-15 feet, so this should be a reasonable benchmark for simulation testing.

The maximum stress, factor of safety (FOS), and maximum deformation are all comparable to Sirius (15,700 PSI, 2.3, and 0.05" respectively). Given that Sirius's UHPV did not experience any issues after being initially leak tested, and survived several significant impacts with the bottom of the pool, it appears that Orion should function similarly.

Table 1: Orion UHPV Analysis Data (15 PSI, approximate pressure submerged at 20 feet)

Max Stress	16,440 psi
Factor of Safety	2.43
Max Deformation	0.046 in

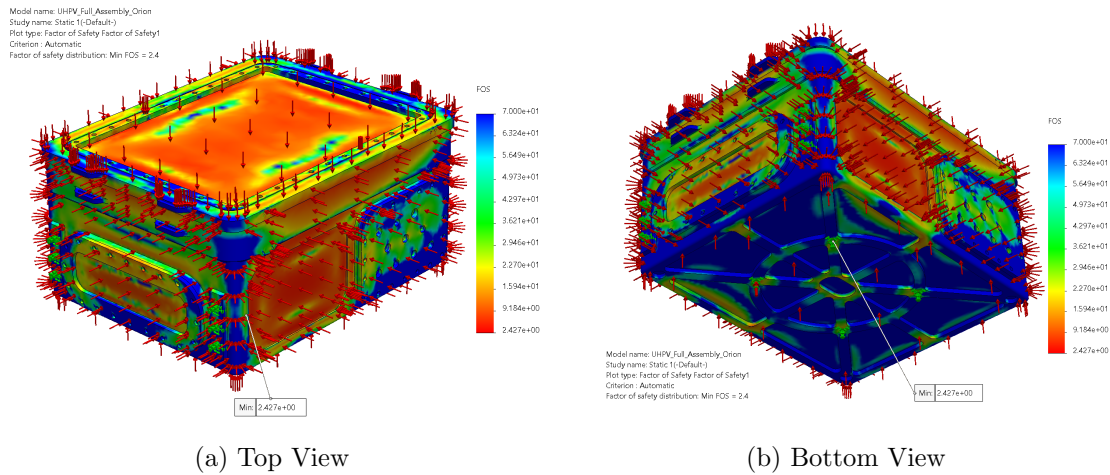


Figure 1: Full Enclosure Factor of Safety (FOS) Simulation Results

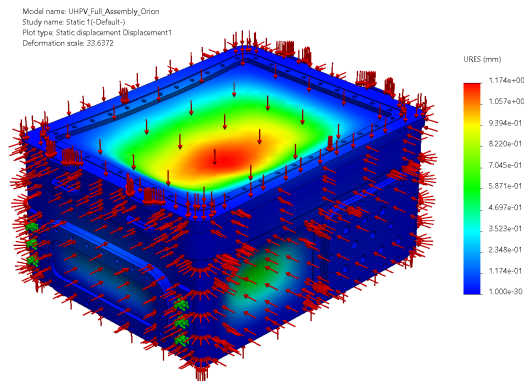


Figure 2: Full Enclosure Deformation Simulation Results

C Additional O-Ring Analysis

For the past few years, we have had some repeated issues with leak testing and sealing surfaces across several UHPVs (more recent issues with Sirius and Polaris come to mind). These have ranged from repeated leaking due to unknown causes, or seals that are difficult to work with and engage properly (especially the large bore seals featured on the UHPVs). As such, I looked more into the Parker O-Ring Handbook (as well as the E-Handbook and other resources) to investigate some of these issues further, and see what potential solutions there are that could address these persistent issues. Some of these solutions were included in my design, but I also found other information and o-ring design principles that we may want to be aware of and test more thoroughly in the future.

The first major set of choices when designing a sealing surface is choosing the o-ring itself. While the diameter of the o-ring is often restricted by the specific part/enclosure being made, the thickness, hardness, and cross-section of the o-ring can all be modified. Outside of the o-ring itself, the o-ring groove can be modified to address certain tolerances and increase ease-of-assembly. Most of these dimensions above have not been discussed much in our previous documentation, so I will break down my findings and analysis on each point.

C.1 Cross-Sectional Diameter

The thickness of an o-ring can be determined by the first digit of its identification number, and ranges from a cross-sectional diameter of 0.070" (for the #0XX or "000-series" o-rings) to 0.275" (400-series o-rings). Typically, we use either 100-series (0.103" diameter) or 200-series (0.139" diameter) o-rings for most of our seals, but the choice to use a specific size has been inconsistent across most enclosures. Generally speaking, the smaller series o-rings are more commonly used for smaller enclosures (and smaller inner or outer diameters for the o-ring), and vice versa. However, there are some cases where using a thicker or thinner o-ring could be particularly useful.

In the case of the UHPV bore seal, it has been difficult to tolerance the o-ring groove (often due to fear of cutting the groove too deep), which has led to a lot of very tight seals. When using thicker o-rings (such as the 300-series with a 0.210" cross-sectional diameter), the tolerance for most dimensions is typically larger, making it easier to machine without fear of completely boning a part past repair. Thicker o-rings are also more resistant to the burring of surfaces (a worsening issue experienced with Polaris), have a longer lifespan, and are more resistant to rolling when being installed (which could compromise the seal). These factors are ultimately why I chose to use the 300-series o-rings for Orion's bore seal (compared to Sirius's 200-series).

The main downside to using a thicker o-ring (beyond the groove and supporting structure needing to be larger) is the additional compressive load that the o-ring applies to the other surface it is contact with. The Parker O-Ring Handbook has several charts which plot the "compressive load per linear inch of seal" depending on a few different factors, such as the squeeze % (how much the o-ring is compressed), the hardness of the o-ring, and its cross-sectional diameter. A 300-series o-ring may exert as much as 2.5 times as much compressional load as a 200-series o-ring, which thus increases the friction required to engage the bore seal appropriately. Ideally, by increasing the tolerance of the o-ring groove and other dimensions, it should be possible to more directly control the squeeze % of the o-ring and reduce this compressional load so that it minimally larger. However, we may not have the precision required to manipulate the squeeze % enough to accurately control this. Working with Orion will help give better insight into this issue.

C.2 Durometer (Shore A Hardness) and O-Ring Shape

The durometer of an o-ring represents how hard (or squishy) the o-ring is, and is measured on the Shore A Hardness scale. The industry standard o-ring has a durometer of 70A, and is what we have been using exclusively up until this point.

In the vast majority of cases, the 70A durometer o-ring is ideal. However, with specifically the large bore seal of the UHPV, it may be beneficial to switch to a softer o-ring (50A) to help address some of the issues outlined above with using a thicker o-ring. Namely, a softer o-ring can reduce the compressive load of the o-ring, reducing the friction generated during installation. They are used most commonly for static, low-pressure (i.e. 1,500 psi) seals, which appears to be our exact use case. Softer o-rings are more susceptible to being pinched or having a shorter lifespan, but these cons seem to be offset by choosing a larger cross-section o-ring. The nice benefit of testing a softer o-ring is that they can be retrofit and are interchangeable in o-ring grooves with the same number, so we can try them out on any current or past design. McMaster does not offer as many softer o-rings as stock (such as the large size needed for the bore seal), but they can be custom ordered.

O-rings most commonly have a circular cross-section, but many types of o-rings with different cross-sections exist. We have almost exclusively used circular o-rings, with the exception of testing some face seals with square o-rings, which appeared to leak more frequently and were thus discontinued. The Parker O-Ring Handbook barely mentions that non-circular o-rings exist, so information on these have to be sought out elsewhere. Again, in most cases, I believe that the standard option (circular) is best.

Looking at the bore seal once again, I was drawn to "X" cross-sectioned o-rings, which have some potential for being particularly helpful. They are advertised as being ideal for low-pressure static seals where a significantly reduced closure force is ideal, or where rolling of the o-ring is possible. If used correctly, it seems like these "X" o-rings would be better than circular o-rings, at least in theory. If they were improperly assembled (twisted when inserted), it is almost guaranteed that the seal would fail, and due to how often we install and remove the lid (especially at competition, where we may be more frantic), it seems that it is too risky to use.

C.3 Groove/Seal Design

Most of our o-ring groove dimensions and tolerancing follows exactly what is recommended in the Parker O-Ring Handbook, which is generally best practice. A commonly missed design feature I have noticed across multiple bore seals is an inner chamfer on the part receiving the o-rings (the female side of the seal). Parker recommends that this leading-edge chamfer is at a shallow angle (between 10 and 20 degrees) to help properly "shoehorn" the o-ring into its compressed state. If too sharp (such as with Polaris's UHPV, which had its chamfer at 45 degrees), the o-ring will get pinched along the surface and be more difficult to install.

The bore seal of the rectangular UHPV requires some modification to the clearance gap recommended by Parker due to the limitation in our tolerancing when using the CNC. The

radial clearance gap for the bore seal is ideally supposed to be $\tilde{3}$ thou, which is simply not possible. Sirius's lid still required extensive sanding when this gap was increased by an additional 30 thou, although significant warping occurred during the welding process. With Orion, we will see if an additional 20 thou is ideal for a clean fit between the lid and hull without exposing the o-rings too much (which could cause pinching and faster wear).